title: "Cover letter"

Re: Submission of "Calibration-derived decoder discriminability is associated with online P300-speller accuracy, but the fitted mapping does not transport across cohorts" as a Paper

Dear Editors of *Journal of Neural Engineering*,

We submit the manuscript above for consideration as a Paper. A score derived from the calibration block that precedes a P300-speller session has repeatedly been related to that session's online spelling accuracy. Mainsah and colleagues took this furthest in this journal, deriving speller accuracy analytically from a calibration-derived detectability index so that performance could be estimated without extensive online testing (*J Neural Eng* 2016;13:066007). That relationship, and the earlier reports it builds on, were established within the cohort in which they were measured. Whether a mapping fitted in one set of cohorts estimates accuracy in a cohort it has never seen is a separate question, and it is the question that decides whether such a score can be reported anywhere other than where it was developed.

We evaluated this in the public BigP3BCI archive. All 20 documented source studies supplied the shared 16-channel montage; 18 yielded eligible online outcomes, giving 271 participants, 410 sessions, 739 session-condition records and 19,611 character selections, with the four documented ALS cohorts as the originally targeted primary subgroup, fixed before the design was widened beyond them. One source study was withheld from development at a time. The estimand is the accuracy of a randomly chosen character selection, because the mapping is fitted on character-expanded records; pooled estimation error was 0.098 (95% CI 0.091 to 0.107) against 0.146 for a benchmark that estimates every record at the development-set mean, and that error is invariant to the estimand, at 0.096 and 0.097 when sessions and participants are weighted equally instead. The association itself held up. Its point estimate was positive in all 18 withheld cohorts, at the level of individual participants ($r = 0.714$, 95% CI 0.650 to 0.768, $n = 271$), and after centring both variables within cohort, which removes cohort means though not differences in variance, protocol or measurement precision ($r = 0.677$, 0.621 to 0.726).

The fitted mapping did not transport, and the calibration intercept settles that without appeal to a threshold. Pooling the 18 withheld cohorts with participant-clustered standard errors, the between-cohort standard deviation of the intercept was tau = 0.87 (95% CI 0.60 to 1.51) against a pooled intercept of -0.06, giving a 95% interval for a cohort not represented in the archive of -1.97 to 1.85 on the log-odds scale, which is the difference between a mapping that badly understates accuracy and one that badly overstates it. The slope agrees and is the more delicate case, at tau = 0.43 (0.30 to 0.77) against a pooled slope of 1.06, with observed values from 0.185 to 2.185. Both conclusions hold at the lower confidence limit of tau, where the intervals are still -1.39 to 1.26 for the intercept and 0.394 to 1.704 for the slope. I-squared is reported with its own interval as a description only, and no conclusion is drawn from where it falls relative to the conventional 75% threshold, because a cluster-robust variance at 5 to 24 participants per cohort is biased downward in exactly the direction that inflates it. In one cohort the score estimated accuracy less well than the development-mean benchmark and in six less well than that cohort's own mean, and for a cohort not represented in the archive the interval on discrimination reaches down to chance.

One cohort-level feature accounts for much of that spread. The median time a cohort took over each selection, which ran from 7.3 to 50.0 seconds and is the archive's only recoverable trace of the stopping rule, accounted for about two thirds of the between-cohort variance in the calibration slope (0.406 per between-cohort standard deviation, 95% CI 0.181 to 0.632; Holm-corrected $p = 0.012$), leaving a residual between-cohort standard deviation of 0.25 against 0.43. Seven further protocol descriptors, including both available proxies for the size of the speller matrix, accounted for no share distinguishable from zero. This strengthens the transportability argument rather than weakening it: a mapping whose slope depends on the stopping rule still cannot be carried to a site whose stopping rule is unknown, and the archive records no stopping rule for any cohort as a field. What it does change is the status of the residual spread, which is not an irreducible property of the population, so a site that reported its stopping rule alongside a calibration score would remove a substantial part of the uncertainty in a transported estimate.

We believe this carries three implications for this journal's readership. First, it separates two claims that the field has tended to report together: an association that appears in every cohort examined, and a calibrated mapping that does not transport. A calibration score can support ranking sessions within a setting, and can support a data-quality screen, but should not be used to report an expected accuracy elsewhere without local recalibration. Second, the four ALS cohorts analysed alone gave an uncorrected between-cohort slope spread of 0.223, against 0.560 across all 18; those four cohorts were not merely too few to estimate the spread, they were homogeneous and favourable, so the resulting picture was optimistic in a measurable direction. Evaluations built on a small number of cohorts should be read accordingly. Third, excluding records in which more than 20% of calibration epochs were rejected lowered estimation error from 0.101 to 0.087 and the uncorrected slope spread from 0.560 to 0.465, the only analysis in which transportability improved, and that screen is actionable at no cost because the rejection fraction is already computed whenever the score is computed.

We note that the journal's guidance asks that contributions using openly available data provide validation on additional data sets and insight into why performance differs. This study reports no performance improvement. It reports a negative transportability result across 18 independent cohorts, together with an examination of which cohort-level features are and are not associated with it. All analysis code, frozen outputs, and the archive provenance record with checksum verification are available to support editorial assessment.

We thank the editors for their time and consideration. The work has not been submitted elsewhere and is not under consideration by any other journal. A companion manuscript from our group uses the same archive to ask a different question, namely how much of each emitted character selection is determined by a language-model prior rather than by the neural signal; it is being submitted to a different journal, reuses the calibration score reported here as a moderator, and a copy is provided with this submission in keeping with ICMJE guidance on overlapping publications.

Sincerely,

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[Institution, department]

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